

Relating Algebraic Statistics with Graphical Models

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Project Goal

Aim of the Project

Relate **graphical models** (especially DAGs and ancestral graphs) to **abstract algebra**: polynomial ideals and Gröbner bases.

Question

Given two DAGs (or ancestral graphs), decide whether they are **Markov equivalent**: same conditional independence (CI) structure (same skeleton and same v -structures for DAGs [VP91]).

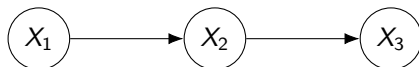
Why it matters?

- **Causal inference**: Structure learning in bioinformatics, vision, NLP, etc.: given data, we often seek the (non-unique) underlying graph.
- **Structure learning**: Fast equivalence tests prune redundant graphs through CI checks.

1. Preliminaries on Graphs

- Motivation: DAGs in Relation to Ring Theory
- Ideals as Algebraic Law

Warm-up: a three-node Markov chain



CI statement: $X_1 \perp X_3 \mid X_2$.

For binary X_i , write

$$p_{ijk} = \mathbb{P}(X_1=i, X_2=j, X_3=k).$$

Quadratic constraint on cell probabilities

$$p_{ijk} \cdot p_{i'jk'} = p_{i'jk} \cdot p_{ijk'}.$$

For binary data this yields explicit binomials f_1, f_2 in the p_{ijk} 's.

Polynomial ring

All p_{ijk} generate a ring $R = \mathbb{F}[\dots, p_{ijk}, \dots]$; CI relations generate an ideal $I = \langle f_1, f_2, \dots \rangle$ of polynomial constraints satisfied on the model.

Each polynomial in I vanishes on all distributions compatible with the graphical CI restrictions.

Ideals: Algebraic “Laws” of the Model

Ideal

A subset $I \subseteq R$ is an **ideal** if $(I, +)$ is a subgroup and $lr, rl \subseteq I$ for all $r \in R$.

Generated ideal

Given $f_1, \dots, f_m \in R$, the ideal generated by the polynomials is:

$$\langle f_1, \dots, f_m \rangle = \left\{ \sum_{j=1}^m g_j f_j : g_j \in R \right\}.$$

Ancestral graph [RS02]

A mixed graph $\mathcal{G}$ in which for all vertices α :

- ① $\alpha \notin \text{ant}(\text{pa}(\alpha) \cup \text{sp}(\alpha))$;
- ② if $\text{ne}(\alpha) \neq \emptyset$, then $\text{pa}(\alpha) \cup \text{sp}(\alpha) = \emptyset$.

Here pa , sp , ne , ant denote parents, spouses, neighbors, and ancestors.

2. Algebraic Structures for Graphs

- Noetherian rings and the Hilbert basis theorem
- The Gröbner Basis

Noetherian rings and finitely many constraints

Noetherian ring

Every ideal is finitely generated; equivalently, the **ascending chain condition** (ACC) holds for ideals.

Hilbert basis theorem

If R is Noetherian, then $R[x_1, \dots, x_n]$ is Noetherian. In particular, every ideal in $\mathbb{F}[x_1, \dots, x_n]$ is finitely generated.

Hence, for the algebraic graphical model (or mixture over a finite parametrization) admits a **finite** polynomial description—the setting where Gröbner methods apply.

Term orders, initial ideals, Gröbner bases

Term order (monomials)

A total order $\prec$ on monomials is a term order if: (i) it is a well-order, and (ii) $x^a \prec x^b \Rightarrow x^{a+c} \prec x^{b+c}$ for all $c \in \mathbb{N}^n$.

Leading data

Fix a monomial order $\prec$ on $\mathbb{F}[x_1, \dots, x_n]$. For nonzero f , $\text{LT}(f)$ is the largest monomial term; $\text{in}_\prec(I) = \langle \text{LT}(f) : f \in I \rangle$.

Gröbner basis

Finite $G \subseteq I$ is a Gröbner basis of I if $\langle \text{LT}(g) : g \in G \rangle = \langle \text{LT}(f) : f \in I \rangle$.

Markov equivalence

Graphs G_1, G_2 on the same variables are Markov equivalent if they imply the same CI statements: $\mathcal{I}(G_1) = \mathcal{I}(G_2)$ (via d -separation on DAGs or m -separation on ancestral graphs).

3. Equivalent Classes of DAGs

- Equivalence of DAGs
- Buchberger's Criterion
- Multivariate Division Algorithm
- Buchberger's Algorithm
- Example of DAGs with Latent Variables

Verma–Pearl characterization for DAGs

Uncoupled head-to-head / Collider (v -structure)

A collider $i \rightarrow k \leftarrow j$ with *no* edge $i \text{---} j$ is an uncoupled head-to-head configuration.

Theorem (Verma–Pearl [VP91])

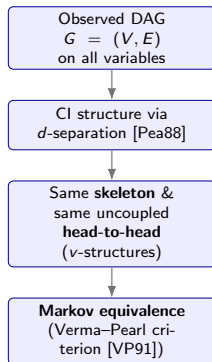
Two DAGs are Markov equivalent iff they share the same **skeleton** (adjacencies ignoring direction) and the same set of **uncoupled head-to-head** nodes.

Limitation for latent settings

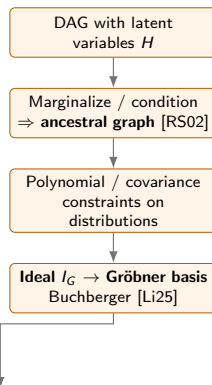
For marginalizations producing **ancestral** mixed graphs, there is no simple graphical checking criteria; m -separation can distinguish graphs that share skeleton and apparent colliders. Algebraic ideals capture finer covariance structure.

Workflow comparison (fully observed vs. latent)

Fully observed DAGs



Latent / marginal setting



Compare two models: graphical same skeleton/ v -structure test
|| ideal / reduced Gröbner basis equality (fixed monomial order)

*Fully observed: purely graphical Markov equivalence.
Latent: polynomials, ideals, varieties, and Gröbner bases (write-up).*

Buchberger's criterion and S -polynomials

S -polynomial

For polynomials f, g with fixed monomial order,

$$S(f, g) = \frac{\text{LT}(g)}{\gcd(\text{LT}(f), \text{LT}(g))} f - \frac{\text{LT}(f)}{\gcd(\text{LT}(f), \text{LT}(g))} g.$$

Buchberger's criterion

$G = \{g_1, \dots, g_k\}$ is a Gröbner basis of I iff every $S(g_i, g_j)$ reduces to remainder 0 on division by G (multivariate division algorithm).

Multivariate Division Algorithm

```
Input:  $G = \{g_1, \dots, g_k\}$ ,  $f$   
 $q_i \leftarrow 0$  for  $i = 1, \dots, k$ ;  $p \leftarrow f$ ;  $r \leftarrow 0$   
while  $p \neq 0$  do  
   $t \leftarrow \text{LT}(p)$ ;  $\text{reduced} \leftarrow \text{false}$   
  for  $i = 1$  to  $k$  do  
    if  $\text{LM}(g_i) \mid \text{LM}(t)$  then  
       $m \leftarrow t / \text{LT}(g_i)$   
       $q_i \leftarrow q_i + m$   
       $p \leftarrow p - mg_i$   
       $\text{reduced} \leftarrow \text{true}$ ; break  
    end if  
  end for  
  if not  $\text{reduced}$  then  
     $r \leftarrow r + t$ ;  $p \leftarrow p - t$   
  end if  
end while  
return  $(q_1, \dots, q_k, r)$ 
```

The multivariate division algorithm terminates by well-ordering of order.

Buchberger's Algorithm

Input: generating set F of ideal I

$G \leftarrow F$

while G fails Buchberger's criterion **do**

 choose $f, g \in G$

$h \leftarrow \text{NF}_G(S(f, g))$

if $h \neq 0$ **then**

$G \leftarrow G \cup \{h\}$

end if

end while

(optional) reduce/simplify G

return G

Theorem (Buchberger's algorithm)

Buchberger's algorithm terminates.

Variables of Markov Graphs

Here, we give an example to show how our approach can identify two Markov equivalent graphs.

$$\begin{aligned}H_1 &\rightarrow A, & H_1 &\rightarrow B, \\H_2 &\rightarrow C, & H_2 &\rightarrow D, \\A &\rightarrow E, & B &\rightarrow E, \\C &\rightarrow F, & D &\rightarrow F, & E &\rightarrow F.\end{aligned}$$

Variable	Role
H_1, H_2	Latent genetic / environmental factors
A, B	Inflammation and immune markers
C, D	Lung and metabolic markers
E	Chronic disease severity
F	Hospitalization risk

Latent DAG and Marginal Ancestral Graphs

Latent biomarker-style DAG

Latents H_1, H_2 drive observed markers $A, \dots, F$ with a fixed directed structure; marginalizing H_1, H_2 yields bidirected edges $A \leftrightarrow B$ and $C \leftrightarrow D$.

$$\begin{array}{ll} A \leftrightarrow B, & C \leftrightarrow D \\ A \rightarrow E, & B \rightarrow E, \\ C \rightarrow F, & D \rightarrow F, \quad E \rightarrow F. \end{array} \qquad \begin{array}{ll} A \leftrightarrow B, & C \leftrightarrow D, \\ A \rightarrow E, & B \rightarrow E, \\ C \rightarrow F, & D \leftrightarrow F, \quad E \rightarrow F. \end{array}$$

The two graphs differ by $D \rightarrow F$ or $D \leftrightarrow F$. They look close graphically, but induce different algebraic constraints after marginalization.

Goal: Test whether the two graphs define the same model ideal by computing reduced Gröbner bases with a fixed monomial order.

Observed output

- Reduced Gröbner basis of J_l : 21 generators.
- Reduced Gröbner basis of J_r : 21 generators.
- Notebook verdict: Do J_l and J_r define the same ideal? `False`.

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